

EX.NO : 10

IMPLEMENTING ERROR DETECTING CODE USING CHECKSUM

AIM :

To write a Java program for Error Detecting code using Checksum.

ALGORITHM :

- 1) Start the program.
- 2) Prompt the user for the number of data segments and the number of bits per segment.
- 3) Read and store the data segments.
- 4) Calculate the sender's checksum and its complement.
- 5) Validate the received checksum and print the conclusion.
- 6) Stop the program.

PROGRAM :

```
import java.util.Scanner;

public class Checksum
{
    static String complement(String sum, int m)
    {
        char bits[] = sum.toCharArray();
        for (int i = 0; i < m; i++)
            {if (bits[i] == '1')
```

```
        {
            bits[i] = '0';
        }
        else
        {
            bits[i] = '1';
        }
    }
    return new String(bits);
}

static String calChecksum(String data[], int k, int m)
{
    int a = Integer.parseInt(data[0], 2);
    int b = 0;
    int c = 0;
    for (int i = 1; i < k; i++)
    {
        b = Integer.parseInt(data[i], 2);
        c = a + b;
        String temp = Integer.toBinaryString(c);
        if (temp.length() > m)
        {
            temp = temp.substring(1);
            c = Integer.parseInt(temp, 2);
        }
    }
}
```

```

        c = c + 1;
    }

    a = c;
}

String sum = Integer.toBinaryString(c);
String t = sum;
if (sum.length() < m)
{
    int diff = m - sum.length();
    for (int i = 0; i < diff; i++)
        t = "0" + t;
}

sum = t;
return sum;
}

static boolean validateChecksum(String data[], int k, int m, String
senderChecksum)
{
    String sum = calChecksum(data, k, m);
    int s = Integer.parseInt(sum, 2);
    int sc = Integer.parseInt(senderChecksum, 2);
    s = s + sc;

    String finalSum = complement(Integer.toBinaryString(s), m);

    System.out.println("Receiver side sum: " +
Integer.toBinaryString(s));

```

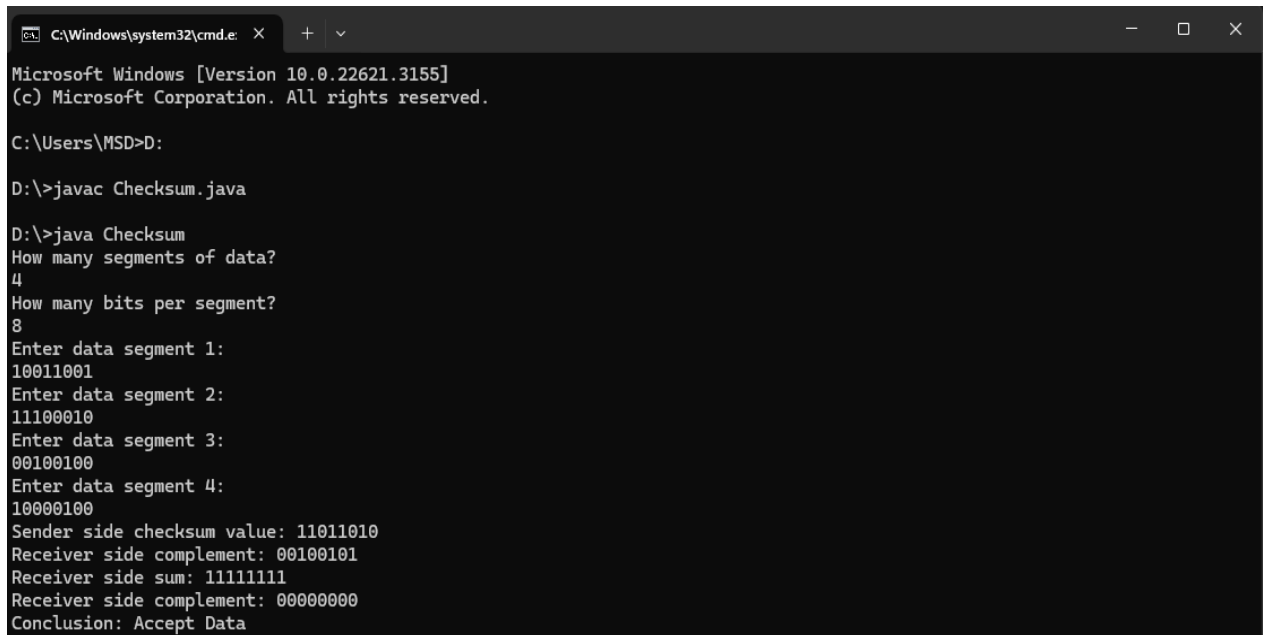
```

        System.out.println("Receiver side complement: " + finalSum);
        return finalSum.equals("00000000");
    }

    public static void main(String[] args)
    {
        System.out.println("How many segments of data? ");
        Scanner input = new Scanner(System.in);
        int k = input.nextInt();
        System.out.println("How many bits per segment? ");
        int m = input.nextInt();
        String data[] = new String[k];
        for (int i = 0; i < k; i++)
        {
            System.out.println("Enter data segment " + (i + 1) + ": ");
            data[i] = input.next();
        }
        String senderChecksum = complement(calChecksum(data, k, m), m);
        System.out.println("Sender side checksum value: " + senderChecksum);
        System.out.println("Receiver side complement: " +
        complement(senderChecksum, m));
        System.out.println("Conclusion: " + (validateChecksum(data, k, m,
        senderChecksum) ? "Accept Data" : "Reject Data"));
    }
}

```

OUTPUT :



```
C:\Windows\system32\cmd.exe X + v
Microsoft Windows [Version 10.0.22621.3155]
(c) Microsoft Corporation. All rights reserved.

C:\Users\MSD>D:

D:\>javac Checksum.java

D:\>java Checksum
How many segments of data?
4
How many bits per segment?
8
Enter data segment 1:
10011001
Enter data segment 2:
11100010
Enter data segment 3:
00100100
Enter data segment 4:
10000100
Sender side checksum value: 11011010
Receiver side complement: 00100101
Receiver side sum: 11111111
Receiver side complement: 00000000
Conclusion: Accept Data
```

RESULT :

Thus, the program for Error Detecting code using Checksum is successfully executed and the output is verified.